

Project Proposal

Bangladesh Constitution Assistant: A Transformer-Based Legal Question Answering System

Submitted To:

Razorshi Prozzwal Talukder

Lecturer

North East University Bangladesh

Course Title: Deep Learning

Course Code: CSE-460

1. Introduction

The rapid advancement of deep learning and natural language processing (NLP) has enabled the development of intelligent systems capable of understanding and generating human-like responses. Legal information systems can benefit significantly from these technologies by providing users with accurate and accessible information.

This project proposes the development of a **Bangladesh Constitution Assistant**, an AI-based question-answering system designed to provide constitutional information to users. The system uses transformer-based deep learning models, semantic retrieval techniques, and a multi-stage processing pipeline to answer questions related to the Constitution of Bangladesh.

2. Project Objective

The primary objective of this project is to develop an intelligent assistant that can:

- Identify whether a user's question is related to the Constitution of Bangladesh.
 - Retrieve the most relevant constitutional article as context.
 - Generate accurate answers using a fine-tuned transformer model.
 - Reduce irrelevant responses and improve answer reliability through a structured NLP pipeline.
-

3. Proposed Methodology

The proposed system consists of three major stages:

3.1 Question Relevance Classification

Initially, every user query is passed through a relevance detection module to determine whether it is a constitutional question or not.

A binary labeled dataset is used for training:

- **Label 1:** Relevant constitutional questions
- **Label 0:** Irrelevant questions

The classification system uses:

- **Sentence Transformer:** `intfloat/e5-base-v2`
- **Vector Database/Search:** FAISS similarity search index

The question embedding is compared against stored embeddings to determine its relevance.

If the output is:

- **0:** The question is considered irrelevant and is not forwarded to the answer generation model.
 - **1:** The question proceeds to the next stage.
-

3.2 Constitutional Article Retrieval

For relevant questions, the system identifies the appropriate constitutional article that can provide context for answering the query.

The retrieval process follows two approaches:

Direct Article Matching:

If the user explicitly mentions an article number, such as "Explain Article 27", the system directly retrieves the corresponding article text from the constitutional article database.

Semantic Article Matching:

If no article number is mentioned, a semantic matching model is used to identify the most relevant article.

The model used:

- **Model:** Fine-tuned `xlm-roberta-base`

This stage ensures that the answer generation model receives appropriate constitutional context.

3.3 Answer Generation Using mBART

The final response is generated using a fine-tuned mBART transformer model.

The model is trained on a Bangladesh Constitution Question Answering dataset containing:

- Constitutional article text as context
- User question
- Expected answer

The input format is:

Context: Constitutional Article Text

Question: User Question

The model generates a constitutional answer based on the provided context.

4. System Architecture

User Question

|

v

Next.js Frontend

|

v

Backend Server

|

v

Relevance Detection

(E5-base-v2 + FAISS)

|

|

|----- Label 0

|

|

|----- Reject Query

|

|

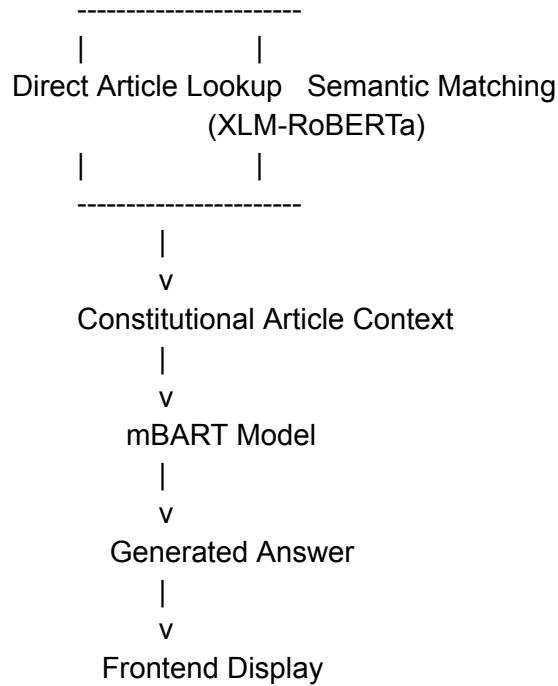
|----- Label 1

|

v

Article Retrieval System

|



5. Technology Stack

Frontend

- Next.js
- User interface for submitting questions and displaying generated answers

Backend

- Python-based backend server
- Google Colab environment for model execution
- Ngrok for exposing the backend service

Deep Learning Models

- Sentence Transformer ([intfloat/e5-base-v2](#)) for semantic representation
- FAISS for vector similarity search
- Fine-tuned XLM-RoBERTa for article matching
- Fine-tuned mBART for answer generation

6. Expected Contribution

This project demonstrates the application of deep learning techniques in the legal information domain. The proposed system integrates:

- Transformer-based language understanding
- Semantic search
- Context-based answer generation
- Domain-specific fine-tuning

The system aims to provide a reliable and efficient method for accessing constitutional information of Bangladesh.

7. Conclusion

The Bangladesh Constitution Assistant combines multiple deep learning approaches to build an intelligent legal question-answering system. By filtering irrelevant queries, retrieving appropriate constitutional context, and generating responses through a fine-tuned mBART model, the system aims to provide accurate and meaningful constitutional assistance.

This project demonstrates the practical application of deep learning, NLP, and transformer architectures in solving real-world information retrieval and question-answering problems.